

BRIGHTPATH LOGISTICS

Autonomous SME Cashflow Early Warning System

Project Story, Scenario Details, Workflow Assignments and Implementation

Specification.

Group 13 | TechCrush AI Automation Capstone.

Cohort 9.

Chosen company: BrightPath Logistics Ltd

Company ID: BP-001

Location: Port Harcourt, Rivers State, Nigeria

Crisis scenario: Fuel price spike

**All monetary amounts are in Nigerian naira (N). Amounts shown as “m” represent millions.
Day 0 is the day the system starts its first forecast.**

DOCUMENT SCOPE

This document defines the BrightPath business story, baseline cashflow, fuel-price crisis, risk rules, AI response, human approval path, reliability scenario, requirements coverage, sample data, n8n workflow architecture, ownership, review responsibilities, and hand-over requirements.

1. The Story of BrightPath Logistics

BrightPath Logistics Ltd is a delivery company based in Port Harcourt, Rivers State. It was founded in 2019 by Mr. Tamuno Briggs, a former warehouse supervisor who saw that many businesses in the city could not find a reliable delivery partner. He began with two second-hand vans and one driver. Today BrightPath runs 12 vehicles (8 vans and 4 light trucks) with 12 drivers and a small office team, delivering goods across the Niger Delta for five main corporate clients.

The business is healthy on paper. It wins contracts, its vehicles are busy and its monthly profit is positive. The weakness is timing. BrightPath pays for fuel and driver wages every week, but its corporate clients pay 45 to 60 days after each invoice. This means the company always funds about seven weeks of operations before it is repaid. Whether it survives a bad month depends on how much cash it holds and on when each payment lands.

Today Mr. Briggs checks the bank balance each morning. That tells him where cash is today, but not where it is heading. Mrs. Ebiere Dokubo, the finance and admin officer, keeps invoices and bills in spreadsheets. Nobody forecasts the next 30 days. BrightPath needs a system that does this automatically and speaks up early.

Company profile at a glance

Item	Detail
Company	BrightPath Logistics Ltd (company_id: BP-001)
Location	Port Harcourt, Rivers State, Nigeria. Founded 2019.
Business	Regional delivery and haulage for corporate clients
Fleet	12 vehicles: 8 vans and 4 light trucks. 3 of the trucks are on a finance loan.
People	12 drivers (paid weekly), 3 office staff and 1 in-house mechanic (paid monthly). 16 in total.
Founder and MD	Mr. Tamuno Briggs (main approver for serious cash decisions)
Finance and Admin Officer	Mrs. Ebiere Dokubo (prepares invoices and bills)
Operations Manager	Mr. Chidi Nwankwo (backup approver if the founder does not respond)
Fuel supplier	Harbour Fuels Ltd, paid weekly for diesel
Fuel use	About 2,880 litres of diesel per week (240 litres per vehicle per week)
Normal diesel price	N1,250 per litre, so a normal weekly fuel bill of N3.6m
Safety buffer	N4.0m, about one week of running costs. The owner wants cash to stay above this.

The five corporate clients

Client	Business	Payment terms	Open invoice
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Coastal Beverages Ltd	Beverage distribution	45 days	INV-1041: N6.4m, due Day 15
Delta Build Supplies	Building materials	60 days	INV-1046: N4.8m, due Day 19
FreshMart Stores	Supermarket chain	45 days	INV-1052: N3.9m, due Day 23
Niger Basin Energy Services	Oilfield services	60 days	INV-1058: N8.5m, due Day 27
QuickCart Nigeria	E-commerce fulfilment	45 days	INV-1063: N3.2m, due Day 30

Total expected from clients over the next 30 days: N26.8m.

2. A Normal Month at BrightPath

In a normal month BrightPath earns about N26.8m in client payments and spends about N21.5m, leaving a profit of about N5.3m. The table shows where the money goes.

Cost	When it is paid	Amount	Monthly total
Diesel (Harbour Fuels)	Weekly, Days 7, 14, 21, 28	N3.6m each	N14.4m
Driver wages	Weekly, Days 5, 12, 19, 26	N0.48m each	N1.92m
Office and mechanic salaries	Monthly, Day 28	N1.0m	N1.0m
Vehicle maintenance and repairs	Days 9 and 23	N0.6m each	N1.2m
Truck loan repayment	Day 10	N1.5m	N1.5m
Vehicle insurance	Day 16	N0.6m	N0.6m
Office rent	Day 2	N0.45m	N0.45m
Tracking software and other costs	Day 20	N0.42m	N0.42m
Total costs			N21.49m

BrightPath starts Day 0 with N16.0m in its main operating account. With no crisis, the lowest point in the 30 days is N5.29m on Day 14, just after the second fuel payment and one day before Coastal Beverages pays. This is comfortable, so the business is HEALTHY.

3. The Crisis: The Fuel Price Spike

A supply disruption at the depots serving Port Harcourt, together with higher import costs, pushes diesel prices up sharply in the first week. Harbour Fuels informs its customers of three increases in five days.

Day	Diesel price per litre	Rise from normal	Weekly fuel bill	Extra cost per week
Day 0	N1,250	None	N3.60m	None
Day 1	N1,500	+20%	N4.32m	N0.72m
Day 3	N1,750	+40%	N5.04m	N1.44m
Day 5	N2,000	+60%	N5.76m	N2.16m

BrightPath's clients are on fixed-rate delivery contracts, so the extra cost cannot be passed on quickly. At N2,000 per litre, four fuel payments cost N8.64m more than normal. The monthly profit of N5.3m becomes a loss of about N3.35m. The bank balance looks fine on Day 5, at N15.07m, so without a forecast nobody would feel the danger until Day 14, when the fuel payment would leave only N0.97m in the account, less than two days of running costs.

With the Early Warning System, the owner hears about the problem on Day 1, 13 days before the danger point.

The 30 day cash calendar: normal month against the full spike (all amounts in N millions)

Day	Event	Balance, no spike	Balance, diesel at N2,000
0	Opening cash	16.00	16.00
2	Rent paid (0.45)	15.55	15.55
5	Driver wages (0.48)	15.07	15.07
7	Fuel payment (3.60 normal, 5.76 at spike)	11.47	9.31
9	Maintenance (0.6)	10.87	8.71
10	Truck loan (1.5)	9.37	7.21
12	Driver wages (0.48)	8.89	6.73
14	Fuel payment, LOWEST POINT	5.29	0.97
15	Coastal Beverages pays 6.4	11.69	7.37
16	Insurance (0.6)	11.09	6.77
19	Wages (0.48), Delta Build pays 4.8	15.41	11.09
20	Tracking and other costs (0.42)	14.99	10.67
21	Fuel payment	11.39	4.91
23	Maintenance (0.6), FreshMart pays 3.9	14.69	8.21
26	Driver wages (0.48)	14.21	7.73
27	Niger Basin Energy pays 8.5	22.71	16.23
28	Fuel payment, staff salaries (1.0)	18.11	9.47
30	QuickCart pays 3.2	21.31	12.67

4. How the System Responds, Step by Step

Each diesel price rise arrives as a financial event. The system recalculates the 30 day forecast every time, using the new price for all four weekly fuel payments.

Risk rules

Status	Rule (based on the lowest projected balance in the next 30 days)	Response
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HEALTHY	N5.0m or more	No action. Normal daily summary.
WATCH	N3.0m to below N5.0m	Alert email to the founder. Keep monitoring.
AT_RISK	N1.0m to below N3.0m	Urgent alert with AI explanation and recommended actions.
CRITICAL	Below N1.0m, or any negative balance	AI analysis, human approval required, all alerts sent.

A breach means the projected balance falls below the N4.0m safety buffer. The forecast reports `days_to_breach`, the number of days from today until that first happens.

What the system reports at each price level

When	Diesel	Lowest projected balance	Status	Breach (below N4.0m)	Warning time
Day 0	N1,250	N5.29m on Day 14	HEALTHY	None	Not applicable
Day 1	N1,500	N3.85m on Day 14	WATCH	Day 14	13 days
Day 3	N1,750	N2.41m on Day 14	AT_RISK	Day 14	11 days
Day 5	N2,000	N0.97m on Day 14	CRITICAL	Day 14	9 days

The state therefore moves from HEALTHY to WATCH to AT_RISK to CRITICAL, and finally to HUMAN ESCALATION when approval is requested. This is the state management the project demonstrates.

What the AI says at CRITICAL

The AI receives only verified figures from the forecast. It returns a structured answer with a short summary, the cause, the key numbers, an urgency level and three recommended actions. A sample of the expected content:

- **Summary:** Cash is projected to fall to N0.97m on Day 14, well below the N4.0m safety buffer.
- **Cause:** Diesel has risen 60% from N1,250 to N2,000 per litre, adding N2.16m to each weekly fuel payment.
- **Action 1:** Ask Coastal Beverages to pay invoice INV-1041 (N6.4m) by Day 13, offering a 2% early payment discount (N128,000).
- **Action 2:** Ask Harbour Fuels to split the Day 14 payment into two instalments of N2.88m, on Day 14 and Day 17.
- **Action 3:** Apply an 8% temporary fuel surcharge to new invoices and review routes to cut fuel use by 10%.

The human approval

At CRITICAL, the Human Approval workflow emails Mr. Briggs with the situation and the recommended actions, with Approve and Reject options. The workflow pauses and waits. If there is no reply within the wait time (2 hours in real life, 2 minutes in the demo), a reminder is sent and

then the request escalates to Mr. Nwankwo. The decision is recorded in the approvals table.

In the story, Mr. Briggs approves Actions 1 and 2. If Coastal pays on Day 13 with the 2% discount, the Day 14 balance rises from N0.97m to about N7.2m, safely above the buffer.

The live failure scenario

During the demo, the AI service is made to fail on purpose, for example by using a wrong API key in W05 at the CRITICAL stage. The Reliability and Audit workflow then retries the AI call three times, records each failed attempt, and switches to a fallback: a rule-based alert built from the forecast figures, clearly marked as an automated summary because the AI was unavailable. The founder still receives the alert and the approval request. The audit log shows the failures, the fallback and the recovery. The key is then restored and the run repeated to show the normal AI path working again.

5. How BrightPath Covers the TechCrush Requirements

#	Requirement	How BrightPath satisfies it
1	2 triggers	Daily 07:00 schedule for the risk check, and a webhook for incoming events (fuel price updates, invoices paid, bank transactions).
2	3 or more integrations	Supabase, Gmail, an AI API, and a fuel price or data source (a public API or a feed the team simulates).
3	Persistent database	Supabase tables hold transactions, forecasts, risk alerts, approvals and audit logs.
4	AI component	AI explains the risk and recommends actions in a structured format.
5	n8n Code Node	JavaScript in W03 builds the 30 day forecast and in W04 applies the risk rules.
6	Sub workflow	W01 calls W02 to W08 using Execute Workflow.
7	Loop or batch	W03 loops through every future payment and invoice, day by day.
8	Human approval	W06 pauses at CRITICAL until the founder approves or rejects.
9	Retry and error handling	W08 retries failed calls, uses a fallback and escalates.
10	Audit log	Every run, failure, decision and recovery is written to audit_logs.
11	Live failure	The AI service is made to fail during the demo (see Section 4).
12	State management	Risk state moves HEALTHY, WATCH, AT_RISK, CRITICAL, HUMAN ESCALATION.

6. Sample Data for Supabase

These values should be loaded exactly, so that every workflow produces the same figures.

Cash position

company_id	date	balance
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BP-001	Day 0	N16,000,000
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Receivables (money owed to BrightPath)

Invoice	Client	Amount	Due day	Status
INV-1041	Coastal Beverages Ltd	6,400,000	15	Open
INV-1046	Delta Build Supplies	4,800,000	19	Open
INV-1052	FreshMart Stores	3,900,000	23	Open
INV-1058	Niger Basin Energy Services	8,500,000	27	Open
INV-1063	QuickCart Nigeria	3,200,000	30	Open

Payables and commitments (money BrightPath owes)

Item	Type	Amount	Due days
Diesel, Harbour Fuels	Payable (amount depends on price)	2,880 litres x price	7, 14, 21, 28
Driver wages	Commitment	480,000	5, 12, 19, 26
Staff salaries	Commitment	1,000,000	28
Maintenance	Payable	600,000	9, 23
Truck loan	Commitment	1,500,000	10

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Item	Type	Amount	Due days
Insurance	Commitment	600,000	16
Office rent	Commitment	450,000	2
Tracking and other	Payable	420,000	20

Fuel price events (used to trigger the crisis)

Event	Day	Old price	New price
fuel_price_update	1	1,250	1,500
fuel_price_update	3	1,500	1,750
fuel_price_update	5	1,750	2,000

Example webhook payload: event_type fuel_price_update, company_id BP-001, fuel diesel, old_price 1500, new_price 1750, event_id FP-003 (the event_id lets W02 detect duplicates).

Suggested Supabase tables: companies, transactions, cash_positions, receivables, payables, commitments, fuel_price_events, forecasts, risk_alerts, approvals, audit_logs.

7. The Workflows for BrightPath

ID	Workflow	What it does for BrightPath	Input	Output
W01	Main Controller	Receives the schedule or webhook event, creates a run_id, decides the path and calls W02 to W08 in order.	Event or schedule	Run summary and final state
W02	Financial Ingestion	Validates incoming events (fuel price updates, payments, invoices), normalizes them, checks duplicates by event_id and stores them.	Event payload	stored, duplicate or invalid, plus record_id
W03	Forecast Engine	Reads cash, invoices, bills and the latest fuel price. A Code Node loops through each future day and event to build the 30 day balance.	company_id, run_id, horizon_days	forecast_run_id, current_cash, expected_inflows, expected_outflows, minimum_projected_balance, day_of_minimum, days_to_breach
W04	Risk Engine	Applies the risk rules in Section 4 with a Code Node, compares with the previous state and saves the alert.	Forecast results	risk_state, previous_state, state_changed, reason
W05	AI Analysis	Sends verified figures to the AI and returns a checked, structured explanation and three actions.	Forecast and risk facts	summary, cause, actions, urgency
W06	Human Approval	Emails the founder, waits for Approve or Reject, reminds, escalates to the Operations Manager and records the decision.	Risk state CRITICAL, AI actions	decision, decided_by, timestamp
W07	Notifications	Sends Gmail alerts for WATCH and AT_RISK, the approval request for CRITICAL, and a daily summary.	Message type and content	sent yes or no, message_id

ID	Workflow	What it does for BrightPath	Input	Output
W08	Reliability and Audit	Retries failed steps, switches to fallbacks, escalates and writes every important event to audit_logs.	Component, error, run_id	action_taken (retry, fallback or escalate)

Call order: START, W01, W02 (if new data), W03, W04, W05, W06 (if CRITICAL), W07, W08, END. Results stay stored in Supabase for review.

8. Workflow Ownership and Team Assignments

The project has 11 members. Ogunleye Oluwatimilehin Favour is the Documentation Lead and

does not own a workflow. The other 10 members each take one main build responsibility plus one supporting responsibility. Every assignment is sized to about 8 workload units on a simple scale: a full workflow build counts 3 to 6 units depending on difficulty, and supporting duties count 2 to 3 units. Heavier workflows are paired with lighter supporting duties, and lighter workflows with heavier supporting duties, so the total effort is equal.

Member	Reg. number	Main build responsibility	Supporting responsibility	Units
Chisom Okafor (Team Lead)	AIA/2026/TC-8/0 677	W01 Main Controller	Integrator: merges approved workflows into the final n8n environment and leads final review	8
Okoye Ngozi Rosemary (Assistant Team Lead)	AIA/2026/TC-8/0 709	W03 Forecast Engine	Task verification and poll oversight	8
Okonji Brendan	AIA/2026/TC-8/0 217	W02 Financial Ingestion	API and integration research, and the simulated fuel price feed	8
Ogunsanwo Ibrahim Ademola	AIA/2026/TC-8/0 444	W04 Risk Engine	Financial logic lead: thresholds, scenario numbers and the state flow	8
Okorafor Ogbonnaya Kennedy	AIA/2026/TC-8/0 934	W05 AI Analysis	Prompt design, structured output format and validation rules	8
Oguegbu Ogochukwu Felicity	AIA/2026/TC-8/0 781	W06 Human Approval	Slide deck and presentation lead	8
Adeleye Okikiola Emmanuel	AIA/2026/TC-8/0 431	W07 Notifications	GitHub custodian (branches, pull requests, no secrets committed) and cloud migration lead	8
Okwor Onyinyechi Favour	AIA/2026/TC-8/0 852	W08 Reliability and Audit	Design of the live failure scenario with the QA lead	8
Olamide Kehinde	AIA/2026/TC-8/0 459	Database and data lead: Supabase tables and the BrightPath sample data	Support for W02 and W03 data access	8
Okeke Ekene Matthew-Daniel	AIA/2026/TC-8/0 803	QA and testing lead: test cases, integration tests, failure test	Demo rehearsal and requirement checklist	8

Ogunleye Oluwatimilehin Favour (Documentation Lead and Co-Lead)	AIA/2026/TC-8/0 293	No workflow. Project diary and tracking	GitHub documentation, screenshot archive and the final contribution PDF with the Team Lead	Documentation
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Review partners

Each workflow is reviewed by a second member before it is merged, so that everyone takes part in quality control.

Workflow	Owner	Reviewer
W01	Chisom Okafor	Okoye Ngozi Rosemary
W02	Okonji Brendan	Olamide Kehinde
W03	Okoye Ngozi Rosemary	Chisom Okafor
W04	Ogunsanwo Ibrahim Ademola	Okonji Brendan
W05	Okorafor Ogbonnaya Kennedy	Ogunsanwo Ibrahim Ademola
W06	Oguegbu Ogochukwu Felicity	Adeleye Okikiola Emmanuel
W07	Adeleye Okikiola Emmanuel	Oguegbu Ogochukwu Felicity
W08	Okwor Onyinyechi Favour	Okeke Ekene Matthew-Daniel

Hand-over from each workflow owner

- The exported workflow JSON.
- A short explanation of what it does.
- The input it expects and the output it returns.
- The tables, APIs and credentials it depends on, without secret values.
- A simple test case with the expected result, using the BrightPath sample data. Credentials stay inside n8n credential management and are never committed to GitHub. Each member builds and tests on local n8n, then the integrator assembles the approved workflows into the cloud n8n instance before presentation day.